

React Endpoint Documentation

This document outlines the React application's routes, the pages they render, the API endpoints they interact with, and the data flow for each interaction.

Authentication Endpoints

1. Login

- **Page:** `/login` (src/pages/auth/Login.jsx)
- **Views:** Login form with email and password fields.
- **Expected Data (from user):** `email` (string), `password` (string)
- **API Endpoint:** `POST /auth/login`
- **Returned Data (from API):** `token` (string), `user` (object: {id, email, name, etc.})

2. Register

- **Page:** `/register` (src/pages/auth/Register.jsx)
- **Views:** Registration form with fields for name, email, password, and password confirmation.
- **Expected Data (from user):** `name` (string), `email` (string), `password` (string), `passwordConfirm` (string)
- **API Endpoint:** `POST /auth/register`
- **Returned Data (from API):** `message` (string), `user` (object: {id, email, name, etc.})

3. Forgot Password

- **Page:** `/forgot-password` (src/pages/auth/ForgotPassword.jsx)
- **Views:** Form to enter email address for password reset.

- **Expected Data (from user):** email (string)
- **API Endpoint:** POST /auth/forgot-password
- **Returned Data (from API):** message (string)

4. Reset Password

- **Page:** /reset-password (src/pages/auth/ResetPassword.jsx)
- **Views:** Form to enter new password and confirm password, along with a token from the reset link.
- **Expected Data (from user):** token (string), password (string), passwordConfirm (string)
- **API Endpoint:** POST /auth/resetpassword
- **Returned Data (from API):** message (string)

5. User Profile

- **Page:** /profile (src/pages/Profile.jsx)
- **Views:** User's profile information (e.g., name, email).
- **Expected Data (from user):** (for update) name (string), email (string)
- **API Endpoint:** GET /auth/profile (to fetch profile), PUT /users/profile (to update profile)
- **Returned Data (from API):** user (object: {id, email, name, etc.})

Device Management Endpoints

1. Get All Devices

- **Page:** /devices (src/pages/Devices.jsx)
- **Views:** A list of all registered devices.

- **Expected Data (from user):** (Optional) Query parameters for filtering/pagination.
- **API Endpoint:** `GET /devices`
- **Returned Data (from API):** Array of `device` objects (e.g., `{id, name, category, purchaseDate, warrantyEndDate, status, etc.}`)

2. Get Device by ID

- **Page:** `/devices/:id` (src/pages/DeviceDetail.jsx)
- **Views:** Detailed information about a specific device.
- **Expected Data (from user):** `id` (string/number) - device ID from URL parameter.
- **API Endpoint:** `GET /devices/:id`
- **Returned Data (from API):** Single `device` object.

3. Create Device

- **Page:** `/devices/new` (src/pages/AddDevice.jsx)
- **Views:** Form to add a new device.
- **Expected Data (from user):** `deviceData` (object: {name, category, purchaseDate, warrantyEndDate, etc.})
- **API Endpoint:** `POST /devices`
- **Returned Data (from API):** Newly created `device` object.

4. Update Device

- **Page:** `/devices/:id` (src/pages/DeviceDetail.jsx)
- **Views:** Form to edit an existing device.
- **Expected Data (from user):** `id` (string/number) - device ID from URL parameter, `deviceData` (object: {name, category, purchaseDate, warrantyEndDate, etc.})

- **API Endpoint:** PUT /devices/:id
- **Returned Data (from API):** Updated device object.

5. Delete Device

- **Page:** /devices (src/pages/Devices.jsx) or /devices/:id (src/pages/DeviceDetail.jsx)
- **Views:** Triggered from device list or detail page.
- **Expected Data (from user):** id (string/number) - device ID.
- **API Endpoint:** DELETE /devices/:id
- **Returned Data (from API):** message (string) - confirmation of deletion.

6. Get Device Categories

- **Page:** Used internally by forms (e.g., AddDevice.jsx, DeviceDetail.jsx).
- **Views:** Dropdown or selection list of device categories.
- **Expected Data (from user):** None.
- **API Endpoint:** GET /devices/categories
- **Returned Data (from API):** Array of category strings (e.g., ["Electronics", "Home Appliances", "Vehicles"])

7. Search Devices

- **Page:** /devices (src/pages/Devices.jsx)
- **Views:** Filtered list of devices based on search query.
- **Expected Data (from user):** query (string) - search term.
- **API Endpoint:** GET /devices/search
- **Returned Data (from API):** Array of device objects matching the query.

8. Get Device Stats

- **Page:** `/dashboard` (`src/pages/Dashboard.jsx`)
- **Views:** Statistics related to devices (e.g., total devices, expiring warranties).
- **Expected Data (from user):** None.
- **API Endpoint:** `GET /devices/stats`
- **Returned Data (from API):** `stats` object (e.g., `{totalDevices, activeWarranties, expiredWarranties, upcomingExpirations}`)

9. Update Device Status

- **Page:** `/devices/:id` (`src/pages/DeviceDetail.jsx`)
- **Views:** Update device status (e.g., active, archived).
- **Expected Data (from user):** `id` (string/number) - device ID, `status` (string).
- **API Endpoint:** `PATCH /devices/:id/status`
- **Returned Data (from API):** Updated `device` object.

10. Upload Document

- **Page:** `/devices/:id` (`src/pages/DeviceDetail.jsx`)
- **Views:** Interface to upload warranty documents or receipts.
- **Expected Data (from user):** `deviceId` (string/number), `file` (File object).
- **API Endpoint:** `POST /devices/:id/documents`
- **Returned Data (from API):** `document` object (e.g., `{id, filename, url, uploadDate}`)

11. Get Documents

- **Page:** `/devices/:id` (`src/pages/DeviceDetail.jsx`)

- **Views:** List of documents associated with a device.
- **Expected Data (from user):** `deviceId` (string/number).
- **API Endpoint:** `GET /devices/:id/documents`
- **Returned Data (from API):** Array of `document` objects.

12. Download Document

- **Page:** `/devices/:id` (src/pages/DeviceDetail.jsx)
- **Views:** Initiates download of a specific document.
- **Expected Data (from user):** `deviceId` (string/number), `documentId` (string/number).
- **API Endpoint:** `GET /devices/:id/documents/:documentId`
- **Returned Data (from API):** Binary file data.

13. Delete Document

- **Page:** `/devices/:id` (src/pages/DeviceDetail.jsx)
- **Views:** Deletes a specific document.
- **Expected Data (from user):** `deviceId` (string/number), `documentId` (string/number).
- **API Endpoint:** `DELETE /devices/:id/documents/:documentId`
- **Returned Data (from API):** `message` (string) - confirmation of deletion.

Maintenance Management Endpoints

1. Get Maintenance Records

- **Page:** `/devices/:id` (src/pages/DeviceDetail.jsx)
- **Views:** List of maintenance records for a device.

- **Expected Data (from user):** `deviceId` (string/number).
- **API Endpoint:** `GET /devices/:id/maintenance`
- **Returned Data (from API):** Array of `maintenanceRecord` objects (e.g., `{id, date, description, cost}`)

2. Add Maintenance Record

- **Page:** `/devices/:id` (src/pages/DeviceDetail.jsx)
- **Views:** Form to add a new maintenance record.
- **Expected Data (from user):** `deviceId` (string/number), `recordData` (object: {date, description, cost, etc.})
- **API Endpoint:** `POST /devices/:id/maintenance`
- **Returned Data (from API):** Newly created `maintenanceRecord` object.

3. Update Maintenance Record

- **Page:** `/devices/:id` (src/pages/DeviceDetail.jsx)
- **Views:** Form to edit an existing maintenance record.
- **Expected Data (from user):** `deviceId` (string/number), `recordId` (string/number), `recordData` (object: {date, description, cost, etc.})
- **API Endpoint:** `PUT /devices/:id/maintenance/:recordId`
- **Returned Data (from API):** Updated `maintenanceRecord` object.

4. Delete Maintenance Record

- **Page:** `/devices/:id` (src/pages/DeviceDetail.jsx)
- **Views:** Deletes a specific maintenance record.
- **Expected Data (from user):** `deviceId` (string/number), `recordId` (string/number).

- **API Endpoint:** DELETE /devices/:id/maintenance/:recordId
- **Returned Data (from API):** message (string) - confirmation of deletion.